

AQUAGUARD AI – UNDERWATER MARINE DEBRIS SURVEY

AI-Powered Underwater Sensing, Multi-Evidence Verification & Cleanup Intelligence Platform

1. Executive Summary & Remediation Recommendation

Operational Priority: PRIORITY 1 | **Action Objective:** Remediation Assessment

Remediation Protocol: Review verified targets and deploy appropriate cleanup equipment.

Recommended Tools: Standard Diver Mesh Sacks | **Execution Window:** Within 48 hours

2. Survey Metadata & Sensing Modality

Session ID: test_verification_session_001	Survey Date: 2026-09-19T09:30:00
Survey Location: Malvan Marine Sanctuary Sector 4	Survey Depth Layer: 12.0 meters
Sensing Modality: OPTICAL RGB	Spatial Positioning: Lat: 16.0583, Lon: 73.4682

3. Input Quality & Pre-Processing Gate Assessment

Optical Quality Score: 86.5/100 **Visibility Index:** Good **Turbidity:** Low **Gate Status:** READY

Applied Enhancement: Adaptive LAB CLAHE

4. Confirmed Marine Debris Targets (Verified)

Target ID	Category	Detect Conf.	Verify Score	AI Est. Depth	Cleanup Tool
TGT-001	Plastic Bottle	90.0%	92.0/100	2.0m	Mesh Sack
TGT-002	Plastic Bag	90.0%	92.0/100	2.0m	Mesh Sack
TGT-003	Can	90.0%	92.0/100	2.0m	Mesh Sack
TGT-004	Fishing Net	90.0%	92.0/100	2.0m	Mesh Sack

5. Rejected & In-Review Candidates Summary

Total Candidates Evaluated: 6 | **Confirmed:** 4 | **Review Required:** 0 | **Safely Rejected:** 2

Rejection Verification Criteria: Frame border touches, insufficient annular local contrast delta, non-conforming aspect ratio, lack of acoustic shadow support.

6. Multi-Factor Environmental Risk Assessment

Ecological Impact Rating: LOW (Score: 3/10)
Low baseline environmental disruption.

7. DBSCAN Spatial Hotspot Analysis

Hotspot Cluster	Target Count	Dominant Category	Cluster Risk	Recommended Action
Hotspot Alpha (North-Left)	3	Plastic Bottle	HIGH	Deploy ROV with mechanical net cutters / heavy retrieval winch immediately.
Hotspot Beta (South-Right)	1	Fishing Net	CRITICAL	Deploy ROV with mechanical net cutters / heavy retrieval winch immediately.

8. Ranked Cleanup Priority Protocol

Rank	Sector Zone	Priority Level	Intervention Protocol
Priority 1	Sector	MEDIUM	ROV Rotary Cutter Blade deployment
Priority 2	Sector	MEDIUM	Diver suction canister recovery

9. Autonomous Inspection Route & Battery Planning

Mission Status: OPTIMIZED (Simulated Autonomous Inspection Route) | **Total Transit Distance:** 84.3 meters | **Estimated Duration:** 17.8 minutes | **Battery Budget:** 7.5%
Waypoints Sequence: START → TGT-004 → TGT-002 → TGT-001 → TGT-003 → RETURN

10. Target Evidence & AI Optical/Acoustic Reasoning

- **TGT-001: Plastic Bottle** — None
- **TGT-002: Plastic Bag** — None
- **TGT-003: Can** — None

11. 2D Coordinate Sector & Spatial Distribution

Spatial Quadrants:

12. Technical Methodology & Model Disclosures

Inference Engine: AquaGuard Precision Pipeline | **Live Latency:** 42.0 ms
Depth Calibration: Depth readings represent AI-estimated visual perspective layers. Operational ROV deployment requires cross-referencing onboard acoustic altimeters.

13. Limitations & Environmental Compliance Sign-Off

Disclaimer: AquaGuard AI is a decision-support prototype for Smart India Hackathon. It does not replace certified marine salvage inspection divers. All detections are subject to multi-evidence verification.
Compliance Status: ISO-Aligned Marine Ecosystem Assessment • **Authorized By:** AquaGuard AI System